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ABSTRACT

The accelerated weight histogram method is an enhanced sampling technique used to explore free energy landscapes by applying an adaptive bias. The method is general and easy to extend. Herein, we show how it can be used to efficiently sample alchemical transformations, commonly used for, e.g., solvation and binding free energy calculations. We present calculations and convergence of the hydration free energy of testosterone, representing drug-like molecules. We also include methane and ethanol to validate the results. The protocol is easy to use, does not require a careful choice of parameters, and scales well to accessible resources, and the results converge at least as quickly as when using conventional methods. One benefit of the method is that it can easily be combined with other reaction coordinates, such as intermolecular distances.

I. INTRODUCTION

Molecular simulation is a powerful tool for studying and understanding conformations and dynamics of molecules, ranging from simple liquids to complex biomolecules. The sampling in standard simulations is limited to states connected by physical pathways that can be reached within the computational time available. However, differences in the free energy between states of the system, or comparisons of free energies of the interaction between different molecules and their surroundings, are of high interest. The most common applications are the free energy of solvation of a molecule and the free energy of drugs binding to one of different molecules. The results of such computations are important for designing new drugs and improving existing drugs as part of a virtual screening workflow or for lead optimization.[1–6] Both relative free energy differences between, e.g., two different molecules in a medium, by transforming one into the other, and absolute free energy differences between a molecule in a medium and in vacuum are of interest.[7] The medium can be a solvent, a binding site of a protein, or an active site of an enzyme.

Monte Carlo (MC) or molecular dynamics (MD) simulations are commonly used to estimate the difference in free energy between two relevant thermodynamic end states. States that involve different numbers or types of molecules may be connected using unphysical, often referred to as alchemical, pathways in which one or more atoms are transformed to another type or created out of nowhere via a sequence of intermediate states. These intermediate states do not have to be realistic in a chemical sense. The path is arbitrary but needs to smoothly interpolate between the end states so that the neighboring states have large enough phase-space overlap.[8–10] With phase-space overlap, it is meant to what extent representative, highly populated, states in one state are also sampled in the neighboring states and vice versa. More generally, the path could involve also other collective variables or global parameters, such as temperature and pressure. In this work, we focus on the example of changing the intermolecular interactions of one solute molecule to its surrounding water solvent in order to calculate its hydration free energy, i.e., its solvation free energy in water.

Free energy simulations are nontrivial, in that they can be sensitive to input parameters, and there are a number of alternative methods to choose from Refs. 7 and 11. Many of these methods also require multiple simulations to be run, possibly in parallel. The alternatives can be divided into three main classes: equilibrium, non-equilibrium, and extended ensemble simulations.

Equilibrium simulations are performed using discrete, often alchemical, states along the reaction coordinate. Each state is simulated separately, and the thermodynamic variables are not modified during the course of the simulations. Alternatively, the sampling could be done using replica exchange.[12] Provided that there is sufficient overlap between the probability distributions of neighboring states, it will be possible to extract the free energy differences. There are multiple methods to analyze equilibrium simulations, such as Thermodynamic Integration (TI),[13] Free Energy Perturbation (FEP),[14] the Weighted Histogram Analysis Method (WHAM),[15] the Bennett Acceptance Ratio (BAR),[8] and the Multistate Bennett Acceptance Ratio (MBAR).[16] The non-equilibrium simulation methods involve measuring the work required as a thermodynamic variable, λ, changes according to a fixed protocol during the course of a simulation, starting from an equilibrium ensemble. The main principle was reported by Jarzynski[17] and extended by Crooks[18] by also taking the reverse transformation into account. Gapsys *et al.* recently demonstrated the efficiency of focusing the simulations on the alchemical end states and running hundreds of short simulations in the forward and reverse directions deriving the free energy difference using the Crooks fluctuation theorem.[18,19] In most cases, a high number of non-equilibrium simulations need to be run, often in the range of 10s to 100s, depending on the system. Extended ensemble methods, also referred to as expanded ensemble methods, extended ensemble Monte Carlo methods, or generalized ensemble methods, include a variety of techniques in which the parameter λ becomes dynamical, effectively sampling from an artificial non-Boltzmann distribution. By modifying the ensemble, it is possible to more quickly escape local free energy minima to improve the sampling. This can be done by applying an adaptive bias depending on how frequently a specific state has been visited. Examples include methods where there is a random walk among the parameters;[20,21] free energy well filling methods, including metadynamics;[22] and histogram methods, such as the Wang–Landau[23] method and the Accelerated Weight Histogram[24] (AWH) method.

In this work, we adapt the AWH method to alchemical transformations. AWH has been available in GROMACS since the 2018 version, controlling the molecular pulling algorithms[25] by setting the potential and force for selected pull reaction coordinates.[26] Herein, we describe the implementation of using AWH to steer an alchemical reaction coordinate. In addition to having a common general framework for reaction coordinates of various kinds, the AWH method offers several other potential advantages in terms of efficiency, accuracy, and robustness. We have verified that the results agree with published data for a simple test system of methane in water. In addition, we have tested its performance, regarding the simulation time required to reach convergence, when calculating the hydration free energy of ethanol and testosterone (see Fig. 1). We have chosen these molecules to cover a range of complexity and properties, with united-atom methane as a simple Lennard-Jones particle, ethanol representing small polar molecules, and testosterone as a more realistic representative of drug-like molecules. We have also experienced, before this study, that alchemical free energy calculations involving testosterone can take a long time to converge and can be considered a challenging case. This selection of molecules is not meant to be comprehensive but still representative to compound types that may be of general interest. Furthermore, we compare with free energy calculations based on equilibrium simulations

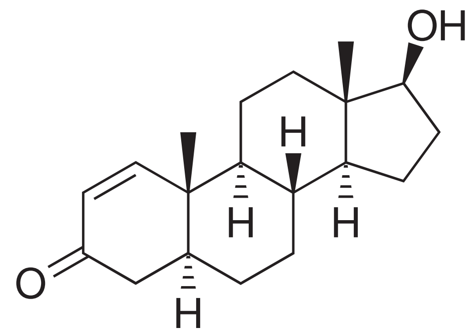

FIG. 1. The molecular structure of testosterone.

analyzed by BAR and MBAR as these methods are popular and have been shown to be reliable.[27] MBAR also makes efficient use of the sampled data.[16] We will show that AWH has the practical advantages that one does not need to fine tune the distribution of λ values and one can obtain the free energy difference directly from a single simulation. It is recommended to choose a relatively large number of λ points. The performance penalty of choosing too many is quite low, as discussed in Sec. II B 3. Even if we herein focus on the basic performance and compare the time of convergence of AWH to commonly used methods, it should be noted that we envisage that the ease of running alchemical free energy calculations as one of the multiple dimensions in a free energy landscape will open up novel research possibilities.

Before discussing our simulations, we give a brief review of the AWH method and how it may be applied to alchemical free energy calculations. We also discuss how both the alchemical path and the sampling distribution along λ may be tuned for additional optimizations and exemplify this with a simple Lennard-Jones fluid.

A. The accelerated weight histogram method

The Accelerated Weight Histogram (AWH) method is a general extended ensemble method, adding an adaptive bias to flatten the free energy landscape in order to improve the sampling efficiency.[24,26] The general idea of extended ensemble methods[20,28] is to simulate samples from a joint probability distribution

$$P(x,\lambda) = \frac{1}{\mathcal{Z}}\pi_\lambda e^{f_\lambda - \beta E_\lambda(x)}, \quad (1)$$

where x denotes the state of the system, e.g., the coordinates and momenta $\{\mathbf{r}^N, \mathbf{p}^N\}$ in an MD simulation and λ represents some parameter(s) of the simulation, which enters the total energy $E_\lambda(x)$ of the system. Typically, we will consider a discrete set of finely spaced parameter values $\lambda \in \{\lambda_k\}_1^M$. $\mathcal{Z} = \sum_\lambda \pi_\lambda e^{f_\lambda - \beta F_\lambda}$ is an undetermined normalization constant. Conditional on λ, the configurations follow an ordinary canonical Boltzmann distribution

$$P(x|\lambda) = e^{\beta F_\lambda - \beta E_\lambda(x)}, \quad (2)$$

where $\beta = 1/k_B T$ is the inverse temperature and $F_\lambda = -k_B T \ln \int e^{-\beta E_\lambda(x)} dx$ is the free energy. In order to adequately sample the parameter values according to a prescribed target distribution

$P(\lambda) \approx \pi_\lambda$, the (hyper) parameters f_λ in Eq. (1) need to be fine-tuned so that f_λ approximately equal the unknown free energies βF_λ. Several different update schemes for f_λ have been proposed in the literature.[20,23,29–33] In AWH, this is accomplished by an adaptive update based on a histogram $W_\lambda = \sum_t w_\lambda(x_t)$ of weights

$$w_\lambda(x) = P(\lambda|x) = \frac{\pi_\lambda e^{f_\lambda - \beta E_\lambda(x)}}{\sum_{\lambda'} \pi_{\lambda'} e^{f_{\lambda'} - \beta E_{\lambda'}(x)}} \tag{3}$$

collected during the simulation.[24] N.B.: since the histograms are sums of fractional weights, their values are not integers.

The extended ensemble (1) can be sampled by alternating between ordinary MC or MD updates at fixed λ and parameter moves keeping x constant. For the latter, it is convenient to employ a Gibbs sampler, drawing λ from Eq. (3). This enables large jumps among the parameter values, which allows for a rather dense spacing with negligible loss of efficiency. Alternatively, one may use a "convolved" update scheme by marginalizing over λ and introducing an effective Hamiltonian

$$H_{\text{eff}}(x) = -k_B T \ln \sum_\lambda \pi_\lambda e^{f_\lambda - \beta E_\lambda(x)} = -k_B T \ln P(x) + \text{const.}, \tag{4}$$

and use this in the MD simulation to generate samples distributed according to the marginal $P(x)$ and then only draw λ from (3) when/if needed.[34]

The AWH algorithm updates the free energy estimates iteratively every ΔN samples (perhaps even with $\Delta N = 1$) by setting for all λ[24,26]

$$f_\lambda \leftarrow f_\lambda - \ln \frac{N\pi_\lambda + \Delta W_\lambda}{N\pi_\lambda + \Delta N \pi_\lambda}, \tag{5}$$

where ΔW_λ is the accumulated weight histogram since the last free energy update. The assumption made here is that the collected weight histogram approximately should follow the marginal distribution of λ, $\Delta W_\lambda \approx \Delta N P(\lambda) = \Delta N \pi_\lambda e^{f_\lambda - \beta F_\lambda}/\mathscr{Z}$, thereby allowing F_λ to be estimated. This approximation will get better and better as the simulation proceeds but may fail at the early stages. The current GROMACS implementation[25,26] uses a two-stage approach where, in the beginning, the (effective) number of samples N is kept constant until a certain covering criterion is fulfilled,[25,26] similar to the iterations in the Wang–Landau method.[23] At each iteration, N is multiplied by a factor > 1, leading to an exponential growth of N and correspondingly an exponential decrease in the update size. This exponential growth stage continues until a second exit criterion is reached, which ensures that the statistical weight of the old samples does not exceed the new ones.[25,26] After this, the algorithm proceeds in the ordinary way, with N increasing linearly with the number of accumulated samples. For large N, Eq. (5) reduces to the asymptotically optimal stochastic approximation based update $f_\lambda \leftarrow f_\lambda - \Delta W_\lambda / N\pi_\lambda$ used also in the self-adjusted mixture sampling method.[31]

It should be noted that the accumulation of the weight histogram ΔW_λ can be done in parallel by running several systems ("walkers"), with common bias f_λ, or by weighting together completely independent AWH simulations. The latter approach allows for straightforward error estimates. When using a common bias, we will see that the efficiency does not depend on the number of walkers. Thus, the number of walkers sharing a bias can be chosen freely to optimize the use of resources and time to obtain solution. The only limit is that with very many walkers, the initial relaxation time might become significant.

1. Alchemical transformations

Specializing to alchemical transformations, the parameter λ from now on interpolates between states with different numbers or types of atoms or molecules as λ varies between, say, 0 and 1. A straightforward way of doing this is by a simple linear rescaling of the interactions $V(\mathbf{r})$ between a particular molecule and the others, e.g., $V_\lambda(\mathbf{r}) = \lambda V(\mathbf{r})$ in the case of particle addition, or $V_\lambda(\mathbf{r}) = \lambda V_A(\mathbf{r}) + (1-\lambda) V_B(\mathbf{r})$ for a transformation between two particle types A and B. In practice, it may be advantageous to consider more complex paths, in which different types of interactions—Lennard-Jones and Coulomb—are turned on sequentially, or more nonlinear modifications to the potentials. For example, when growing in a new particle into a crowded system, one could increase its size or at least soften the short range repulsion. When calculating binding free energies, it may also be helpful to temporarily use, and account for, a restraint potential to hold a ligand in place in the vicinity of a binding site. Importantly, only the end states need be well defined physically realizable states of the system while the intermediate ones need not be.

Working out the optimal path is, in general, a difficult problem. One approach to systematically compare paths consists in computing a "thermodynamic length"

$$\mathscr{L} = \int_0^1 \sqrt{g_\lambda} \mathrm{d}\lambda \tag{6}$$

using an appropriate metric g_λ on the parameter manifold.[10,34–37] For slowly controlled non-equilibrium processes in the quasistatic limit, the relevant metric is the Weinhold metric;[35,36] this was later generalized to (slightly) faster processes, but still within a linear response regime, by accounting for time correlations.[37] The optimization of the thermodynamic length corresponds to the minimization of dissipation in these cases.

For extended ensemble simulations using AWH, where λ is *dynamic*, a metric may instead be defined such that optimal paths minimize the variance of the estimated free energy difference $\overline{\Delta F}$.[34] The latter may be approximated by

$$\text{Var } \beta\overline{\Delta F} \approx \frac{2}{\tau} \int_0^1 \frac{g_\lambda}{\pi_\lambda} \mathrm{d}\lambda, \tag{7}$$

with a metric given by an integrated correlation function of the form

$$g_\lambda = \int_0^\infty \frac{\langle \delta\mathscr{F}(t) \delta\mathscr{F}(0) w_\lambda(x(t)) w_\lambda(x(0)) \rangle}{\langle w_\lambda^2(x(0)) \rangle} \mathrm{d}t \tag{8}$$

(for the case of a one-dimensional parameter manifold, see Ref. 34 for the general case). Here, $\mathscr{F}(t) = -\partial \beta E_\lambda(x(t))/\partial\lambda$ is the thermodynamic force conjugate to λ and $\delta\mathscr{F}(t) = \mathscr{F}(t) - \langle \mathscr{F} \rangle_\lambda$. The variance decreases inversely proportional to time τ as it should. The right-hand side of Eq. (7) is bounded from below by $2\mathscr{L}^2/\tau$, which is attained by choosing the target distribution to be $\pi_\lambda \propto \sqrt{g_\lambda}$. It should be noted that the metric (8) accounts for time correlations in the dynamics, which makes it easy to identify bottlenecks along λ. (This is in contrast to approaches using the Fisher–Rao

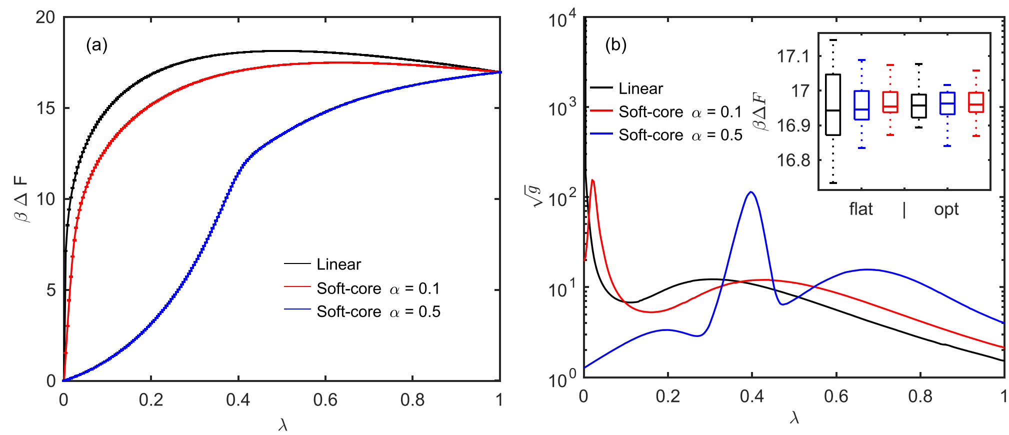

FIG. 2. (a) Free energy difference along λ for different paths and (b) the square root of the metric $\sqrt{g_\lambda}$ along the same paths. The peaks in the metric correspond to regions that are difficult to sample. For the linear scaling, there is a very high and sharp peak at $\lambda = 0$, which can be traced to the divergent energy when Lennard-Jones particles overlap. The soft-core potentials avoid this by regularizing the divergence at the origin and lead to lower and wider peaks. For $\alpha = 0.1$, the peak is still located near the origin, while for $\alpha = 0.5$, it occurs almost half-way to $\lambda = 1$. The inset shows box plots of the distribution of ΔF from 24 independent simulations for the different paths and flat or optimized target distribution. The middle line in each box shows the median, the top and bottom edges represent the upper and lower 25%-quantiles, and the whiskers extend to the maximum and minimum values.

metric,[10,29] where such correlations are neglected.) Choosing the target as $\pi_\lambda \propto \sqrt{g_\lambda}$ will automatically direct the sampling to the difficult regions, which can significantly enhance the sampling.

2. Lennard-Jones fluid example

To illustrate, we briefly consider a two-dimensional Lennard-Jones fluid for a selection of different paths interpolating between $N-1$ and N particles. The free energy difference $\Delta F = F(\lambda = 1) - F(\lambda = 0)$ is directly related to the chemical potential $\mu = F_N - F_{N-1} = \Delta F + k_B T \ln(N\Lambda^2/V)$, where the last ideal gas term comes from the extra "ghost"-particle at $\lambda = 0$ since that still contributes to the kinetic energy compared to the system with $N-1$ particles ($\Lambda = h/\sqrt{2\pi m k_B T}$ is the thermal wavelength). Figure 2(a) shows the free energy as a function of λ calculated using the AWH method along three different paths for an 8×8 system with $N = 64$ Lennard-Jones particles simulated in an NVT ensemble extended along λ, which is discretized into 256 points λ_k. The Lennard-Jones parameters were $\varepsilon = 0.25, \sigma = 1$, and temperature was set to $k_B T = 0.25$. The topmost curve uses a simple linear scaling of the potential, the next uses a soft-core potential with $\alpha = 0.1$, and the lowest curve with $\alpha = 0.5$. The linear scaling is known to be problematic near $\lambda = 0$, where the free energy increases very steeply. A simple modification is to rescale $\lambda \to \lambda^4$ instead, but this does not completely resolve the problem, which comes from the divergent behavior of the Lennard-Jones potential near the origin. Better is to smoothen this divergence by using a soft-core potential,[25,38]

$$V_{sc}(r,\lambda) = 4\varepsilon\lambda\left(A^{-2} - A^{-1}\right), \quad \text{where } A = \alpha(1-\lambda) + (r/\sigma)^6. \quad (9)$$

Once a path has been chosen, it remains to decide how the samples should be distributed along the path. In AWH, this amounts to choosing a target distribution π_λ while keeping the λ-spacing fixed.[39] The most commonly used choice would be a flat distribution, but as discussed above, and in more detail in Ref. 34, the optimal choice is $\pi_\lambda \propto \sqrt{g_\lambda}$. In Fig. 2(b), we show the square root of the metric for the paths in (a). As seen, it varies by orders of magnitude, and therefore, the optimization of π_λ can be expected to help the sampling significantly. The free energy estimates from the simulations using flat and optimized target distributions are listed in Table I together with their standard deviations. Evidently, optimizing the target distribution can reduce errors quite significantly, especially in the challenging case of linear scaling. These estimates were obtained from 24 independent AWH runs weighted together as described in Ref. 24, with the standard errors calculated using bootstrap. A more robust view on the statistical errors can be obtained by forming histograms of the 24 free energy estimates. The inset of Fig. 2(b) shows the corresponding box plots for the different paths and flat or optimized

TABLE I. Free energy differences and their standard errors calculated along three different paths and using flat and optimized target distributions. Note that the statistical uncertainty of the standard errors is about 15%.

	$\beta\Delta F$ flat std. err.	$\beta\Delta F$ optimized std. err.
Linear scaling	16.932 ± 0.023	16.960 ± 0.009
Soft-core $\alpha = 0.5$	16.955 ± 0.013	16.958 ± 0.009
Soft-core $\alpha = 0.1$	16.969 ± 0.010	16.959 ± 0.010

target distribution. The linear scaling has a notably larger error for the flat case, but, perhaps surprisingly, for the optimized case, the error is quite close to that of the other paths. In fact, the errors after optimization are all very similar. The error estimated from Eq. (7) was only accurate to within a factor of 2 but follows the same trend. Since Eq. (7) is based on a Markovian approximation of the dynamics along λ, perfect agreement cannot be expected. It should be noted that the optimal path and target distribution are generally system dependent, which complicates things. On the other hand, the optimization of the target distribution alone might often be sufficient and can, in principle, be done on the fly.[40]

II. METHODS

A. GROMACS implementation of sampling alchemical reaction coordinates using AWH

The AWH implementation in GROMACS was extended to use an alchemical coordinate. This can be used on its own or as one of the multiple reaction coordinate dimensions, where the others are linked, e.g., to the center of mass pulling.[25] AWH is used together with the standard GROMACS FEP user input list of discrete λ states i.e., steps along a reaction coordinate connecting the two end states of interest. It is possible to control Coulomb, van der Waals, bonded, or restraint interactions as well as the temperature of the system. These changes are included in GROMACS 2021.

As mentioned above, AWH allows using either a convolved bias potential or a rejection-free Monte Carlo sampling, also called Gibbs sampling, to apply the bias potential and steer the reaction coordinate. However, currently, only the Gibbs sampling alternative can be used, in GROMACS, if there is an alchemical reaction coordinate dimension since the λ states are clearly defined and discrete. When sampling, the probability $w_\lambda(x)$ of the current configuration visiting all free energy λ states is recorded, and the λ state of the simulation is steered according to the same probability. This allows arbitrarily large steps along the reaction coordinate but with a higher probability of visiting similar states. At regular intervals, the collected sampling data are used for updating the evolving bias and the potential of mean force (PMF). The autocorrelation of $\delta H/\delta\lambda$, i.e., the Hamiltonian derivative, at each λ state, is used to determine the AWH friction metric along the alchemical λ axis.

AWH in GROMACS is not yet capable of automatically optimizing the target sampling distribution based on the AWH friction metric (see Sec. I A 2), but this is planned for future versions.

B. MD simulations

The MD simulations were performed using GROMACS 2021,[41,42] which includes the functionality to use AWH to sample alchemical free energy λ states (see Sec. II A). Temperature was controlled by using a stochastic dynamics integrator (also referred to as a velocity Langevin dynamics integrator) with a time step of 2 fs and with a time constant τ of 2 ps (corresponding to a friction constant of 0.5 ps^{-1}). The pressure (set to 1 atm) was controlled using a Parrinello–Rahman barostat[43] with a time constant of 10 ps and a compressibility of 4.5×10^{-5} bar^{-1}. All bonds to hydrogens were constrained using the LINCS algorithm,[44] and water was kept rigid using SETTLE.[45] TIP3P[46] water parameters were used. Topologies, i.e., inter- and intra-molecular interaction parameters, for ethanol and testosterone were generated using STaGE,[47] which, in turn, uses Open Babel[48] and MATCH.[49]

For ethanol and testosterone simulations, the van der Waals interactions had a cutoff of 1.2 nm with a smooth force switch from 1.0 to 1.2 nm. The methane and ethanol simulations were run with long range dispersion corrections for energy and pressure. The testosterone simulations were originally configured to be compatible with CHARMM force field lipid bilayer simulations in which dispersion corrections should not be used.[50] This does not have any effect on the convergence, which is the focus, herein, of the simulations, but the actual results are affected by the use of dispersion corrections. Our experience from similar testosterone hydration free energy calculations have shown that adding a dispersion correction would lower the hydration free energy by ~8.5 ± 0.3 kJ mol^{-1}. This should be taken into account when comparing these results to experimental data or other simulations.

Coulomb interactions were calculated using Particle mesh Ewald (PME)[51] with a radius of 1.2 nm. $\delta H/\delta\lambda$ and δH to all neighboring λ states were written every 5 ps, except for ethanol expanded ensemble Wang–Landau[20,23,52] simulations in which the $\delta H/\delta\lambda$ output was written every 2 ps. Soft-core transformations[25,38] with α = 0.5 and σ = 0.3 nm [similar to Eq. (9)] were applied to the van der Waals interactions of the solute by setting $V_{sc}(r,\lambda) = \lambda V(r_\lambda)$, $r_\lambda = (\alpha\sigma^6(1-\lambda) + r^6)^{\frac{1}{6}}$, where $V(r)$ is the normal "hard-core" van der Waals potential in the fully interacting state (λ = 1), α is an input soft-core parameter, and σ is the radius of the interaction.

The AWH simulations had a constant, flat target distribution. The GROMACS implementation of AWH uses a two-stage approach with an exponential histogram growth in a short initial stage and a linear growth in the final stage. The histograms were equilibrated as part of the initial stage using the criterion that at least 80% of the target region was sampled within 20% of the target distribution. The length of these initial stages took up only a small fraction of the total AWH simulation times. There were 100 steps between each sampling of the λ-value and 10 samples per update of f_λ. If running multiple bias-sharing walkers, their samples are pooled to compute new weights when updating f_λ, i.e., every 1000 steps in the simulations presented herein. The estimated initial error was 10 kJ mol^{-1}. The initial histogram size, which determines the initial update size of the free energy, is set indirectly by specifying an estimate of the diffusion constant along the λ axis. We used values of the input diffusion constant in the range 4.5×10^{-5} to 4.5×10^{-3} ps^{-1}. This value is not very sensitive and only has a weak effect on the convergence rate, as shown below. These values correspond to diffusing over the whole λ-range in around a nanosecond, which is a reasonable value for many alchemical calculations.

The equilibrium simulations were analyzed using BAR[8] (from GROMACS 2021[41]) and MBAR[16] (alchemical_analysis.py[53] using pymbar 3.0.5[54]). When performing the MBAR analyses, the built-in subsampling of correlated data was used.

A combined result from the different simulations was calculated using the 95% confidence interval of the weighted average of the results at full simulation time. This was obtained by bootstrapping, with 50 000 repetitions, the output from the individual simulations. The weight of each result was its accumulated simulation time, which was almost the same for all ethanol simulations, but we put more emphasis on the extended testosterone equilibrium simulations with 27 λ points. The results with 35 ns equilibration time were also included in the testosterone data. The equilibration time here refers to how much data are discarded at each λ state before starting the analyses controlled by the BAR and MBAR tools. The results from BAR and MBAR analyses were included separately even if they were derived from the same simulation data. For ethanol, the total weight ratio from AWH:equilibration simulations:expanded ensemble Wang–Landau was 4:2:1, meaning that AWH had a higher impact on the combined results. For testosterone, the AWH:equilibration simulations weight ratio was 1:1.25, putting slightly more weight on the equilibrium simulations.

1. Validation—Methane hydration free energy

The methane simulations used previously published parameters.[27,55] In summary, the Coulomb interactions were using PME with a Fourier spacing of 0.12 nm and a potential-switch modifier from 0.88 to 0.9 nm and van der Waals interactions using a potential-switch modifier from 0.8 to 0.9 nm. The temperature was set to 300 K. There were also a few modifications due to differences between GROMACS version 4.0.4 (from 2009) and version 2021 as follows:

- A buffered Verlet pair list was used instead of pair lists based on particle groups.
- A stochastic dynamics integrator was used (also functioning as a thermostat with the time constant, τ, set to 2 ps) instead of the MD velocity Verlet algorithm[56] in combination with a Nosé–Hoover temperature coupling[57,58] with a τ of 5 ps.
- The pressure (set to 1 atm) was controlled using a Parrinello–Rahman barostat[43] instead of a Martyna–Tuckerman–Tobias–Klein (MTTK) barostat.[59]

The free energy of hydration of OPLS-UA (Optimized Potentials for Liquid Simulations-united atom) methane was calculated to verify the results against previously published values[27] and to check consistency between equilibrium simulations and AWH. The reference simulations were run using eight van der Waals λ states—one per simulation—and they were analyzed using BAR. The λ states were at [0.0, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, and 1.0], where at $\lambda = 0$, there is no interaction between the ligand and its surrounding and at $\lambda = 1$, there is full interaction. Six sets of simulations were started in the fully interacting state from equilibration output 1–6 of the AlchemistryWiki[55] dataset. Each subsequent λ state simulation started from the output from the previous state in order to reduce the required equilibration time in each λ state. The simulation time at each λ point was 5 ns, with 500 ps discarded as equilibration. In the original publication,[27] 100 simulations were run, with the same simulated time per λ point and discarded equilibration time as herein. Like in the study by Paliwal and Shirts,[27] three λ states were also selected ([0.0, 0.5, and 1.0]) to allow comparisons to AWH run with only three λ states. In order to retain the same accumulated simulation time, these states were extended to 13.4 ns each.

Six AWH simulations were started from snapshots 1–6 from the equilibration from the AlchemistryWiki.[55] In each AWH simulation, the methane started in the fully interacting state and used the same λ states as in the reference simulation and, in turn, the same as in previously published evaluations.[27] As described above, another batch of six AWH simulations were run with only three λ states. In both of these setups, the AWH simulations were run for 40 ns each.

2. Ethanol hydration free energy

The first convergence test was calculating the hydration free energy of all-atom ethanol. The main difference, compared to UA methane, was that Coulomb interactions were present. The simulations were started from an equilibrated box of water with one ethanol molecule with its interactions to the surroundings completely turned off.

16 λ states were used, distributed as shown in the Appendix, with six states for a linear decoupling of Coulomb interactions, followed by 11 states turning off the van der Waals interactions, with one λ state shared. The equilibrium simulations were first run in a sequence starting at the fully decoupled state. In order to reduce the equilibration time, each λ state was first simulated for 1 ns, starting from the final configuration of the previous simulation (except for the first one, which started from the equilibrated water box with ethanol fully decoupled). After that, the simulations at each λ point were extended to 19 ns and run in parallel. The equilibrium simulations were run in five sets, giving a total simulation time of $5 \times 16 \times 19\,\text{ns}^3 = 1520\,\text{ns}$. The simulations were analyzed using BAR and MBAR.

The AWH simulations all started from the same ethanol configuration in its fully decoupled state. The convergence of the AWH simulations was evaluated using 20 sets of simulations using 1 walker (75 ns each), 5 sets of simulations using 4 communicating walkers (75 ns each), 5 sets of simulations using 16 communicating walkers (19 ns each) and 5 sets of simulations using 48 walkers (6 ns each). The input diffusion constant for the AWH coordinate dimension, which affects the initial AWH update size, was set to $4.5 \times 10^{-4}\,\text{ps}^{-1}$.

After having compared the equilibrium simulation results to AWH, a small discrepancy was found. In order to rule out an implementation error in the AWH functionality in GROMACS, a set of expanded ensemble simulations were performed as a comparison. 20 simulations were run for 75 ns using the Wang–Landau[23] algorithm to determine the weights of the states, i.e., flatten the free energy landscape (determine f_λ), followed by an expanded ensemble stage[20,52] (using the Metropolized Gibbs algorithm for moving between states) with fixed f_λ. The initial Wang–Landau delta was 1.0 k_BT, the criterion for a flat landscape was 0.8, and the scaling factor applied to the delta every time the landscape was flat was 0.7. The equilibration stage (the Wang–Landau update stage) was left when the delta reached 0.001. These simulations were analyzed using MBAR, starting the analyses after leaving the equilibration stage, i.e., when the energy landscape had been flattened. As discussed in more detail below, these results were in full agreement with the AWH simulations.

3. Testosterone hydration free energy

Testosterone was inserted at a random position in a box of equilibrated water and was used as the starting point of the fully decoupled state. The convergence was evaluated using varying number of λ points, both for AWH and equilibrium simulations, and number of communicating AWH walkers as well as the input estimate of the AWH diffusion speed along the λ reaction coordinate.

AWH simulations were run with, a linear set of 16 λ states (see the Appendix), 71 λ states with 51 and 21 (one shared) linearly distributed states for van der Waals and Coulomb interactions, respectively, as well as 141 λ states with 101 and 41 (one shared) states (cf., 16 λ states in the Appendix). 20 simulations with 1 walker (200 ns each), 5 simulations with 4 bias-sharing walkers (200 ns each), 5 simulations with 16 walkers (50 ns each), and 5 simulations with 96 walkers (9 ns each) were tested. The influence of the input diffusion rate along the λ reaction coordinate was evaluated for 16 λ points (4 communicating walkers) with 4.5×10^{-5}, 4.5×10^{-4}, and 4.5×10^{-3} ps^{-1}. This was done to study the effect of the initial update size on the results and convergence. AWH (as well as MBAR) requires the computation of the free energy difference of the current λ state to all other λ states. This results in a slight overhead in simulation times. Testosterone simulations with 141 λ states were approximately one third slower compared to using only 16 λ states. However, this difference is small enough that it is worth adding more λ points if there is any reason to believe that it is needed in order to improve the phase-space overlap between λ states.

Almost the same setup and parameters, as described in Sec. II B 2, were used when running equilibrium simulations for analysis using the Bennett Acceptance Ratio (BAR) and Multistate Bennett Acceptance Ratio (MBAR) methods. The difference was that there was one separate simulation run for each λ state. Two different setups of λ states were employed. In the first case, five sets of 16 λ states were used with the same distribution as in Sec. II B 2 (see also the Appendix). Each set of simulations was started in sequence, as in the ethanol case described in Sec. II B 2, but running 5 ns per λ state instead of 1 ns. After that, each λ state was extended to 50 ns. In the second case, the results from the first 1 ns of 21 linearly distributed λ states (see the Appendix), decoupling Coulomb and van der Waals interactions separately, were analyzed using BAR. This was used to optimize the λ point distribution by placing the points so that the estimated standard deviation per sample between two neighboring λ states was as close to 1.0 k_BT as possible, in order

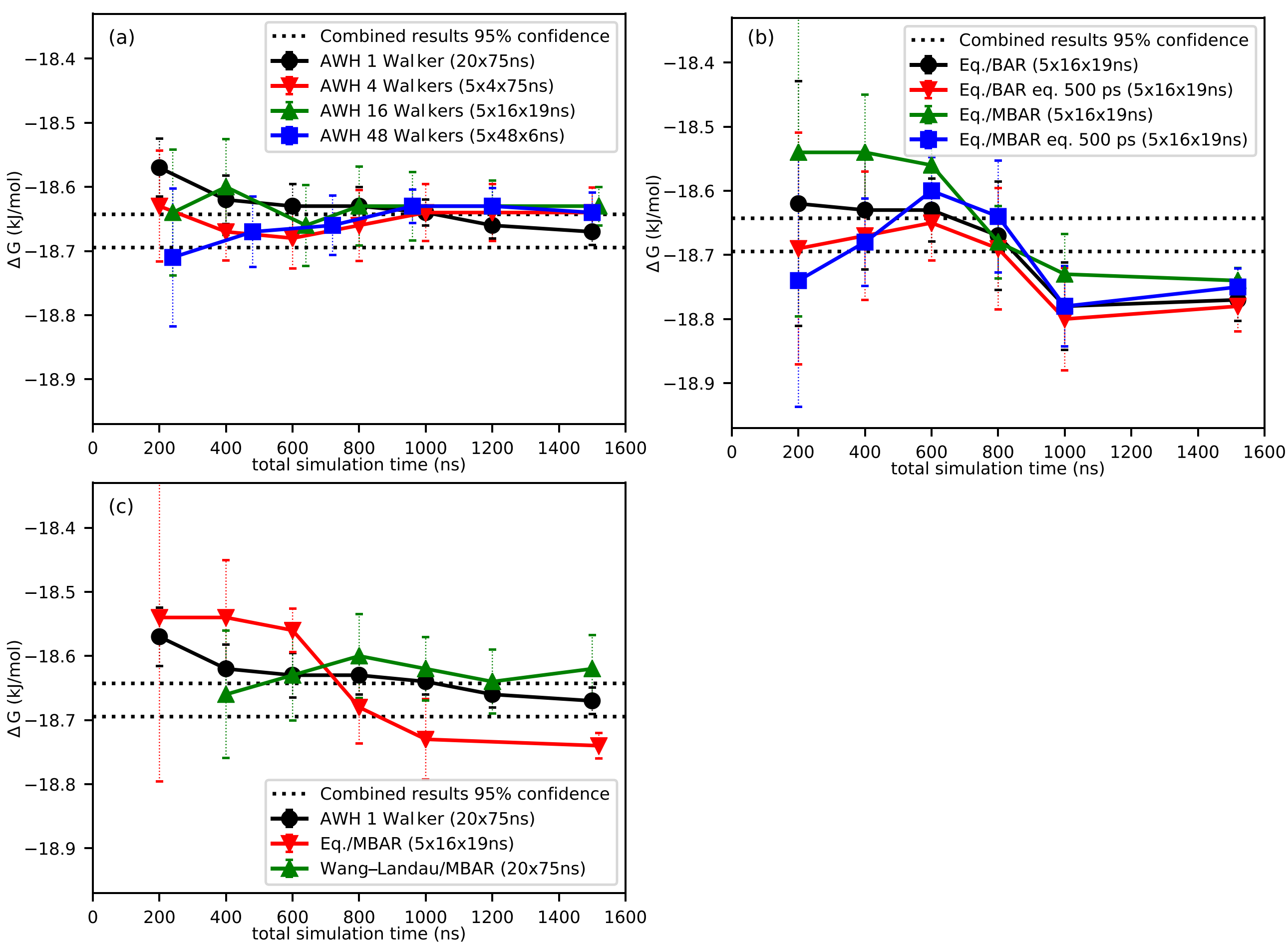

FIG. 3. Convergence of ethanol hydration free energies using AWH, equilibrium simulations (BAR and MBAR) and expanded ensemble Wang–Landau simulations.[20,23,52] The plotted time is the total time of all simulations used for calculating the average free energy. (a) Number of communicating walkers in AWH simulations. (b) Equilibrium simulations analyzed using BAR and MBAR with or without discarding 500 ps as equilibration. (c) Comparing the AWH output to the equilibrium simulation results with the lowest variability after 1500 ns total simulation time (MBAR without equilibration) as well as results from expanded ensemble Wang–Landau simulations analyzed using MBAR. The uncertainties in the plots are 1 SEM.

to improve the efficiency of the sampling.[10] This resulted in 27 λ points with their distribution presented in the Appendix. Five sets of simulations were run for 70 ns at each λ state, starting from the output from the closest (in λ) from the 21 linearly distributed simulations used for optimizing the λ point distribution, for a total simulation time of 9.45 μs (not including the optimization phase). When analyzing, it was also examined how discarding the first 5 ns, or 35 ns, of each simulation would affect the convergence and results.

III. RESULTS

In the analyses of the results below, a level of $p < 0.05$ was used as the threshold for significance. The results presented in Figs. 3 and 5 are the average solvation free energy from five simulations, except for AWH with 1 walker, in which case 20 simulations were used, with the same time per simulation as if running 4 walkers per job. The reported uncertainties represent the standard error of the mean (SEM) from these five (or 20) simulations. It is important to keep in mind that the SEM may be underestimated in case of insufficient sampling. An example of this can be seen in Fig. 5(d), where the BAR results from 16 λ points after 800 ns total simulation time (10 ns per simulation) are significantly different to the results after a total simulation time of 4000 ns (50 ns per simulation). Since only five datasets (in most cases) are used as input, it is also possible that the SEM is underestimated if the results are similar by pure chance.

In Figs. 4 and 6 are shown the root mean square deviation (RMSD) to the average combined results from all full-length simulations.

A. Validation—Methane hydration free energy

There was no significant difference between the calculated average hydration free energy from six sets of equilibrium simulations run at separate λ points analyzed using BAR and from running six AWH simulations. The total simulation time was similar in both cases. Neither the equilibrium simulations using static λ states nor the AWH simulations showed any difference between using eight or three λ states, except that the SEM from the equilibrium simulations was higher with only three λ states.

The calculated methane hydration free energy from six sets of equilibrium simulations of the reference configuration, with eight static λ states run for 5 ns each and analyzed using BAR, was 8.59 ± 0.05 kJ mol^{-1} (with the standard error of the mean from the six simulations as the uncertainty), and the standard deviation was 0.10 kJ mol^{-1}. When analyzing the simulations using only three λ states (extended to 13.4 ns sampling time each), the result was 8.63 ± 0.07 kJ mol^{-1}. The same data analyzed with MBAR gave 8.59 ± 0.06 kJ mol^{-1} (standard deviation of 0.14 kJ mol^{-1}) from all eight λ states and 8.62 ± 0.08 kJ mol^{-1} from three λ states with extended simulation time.

The average hydration free energy from the six AWH simulations was 8.57 ± 0.03 kJ mol^{-1} (with the standard error of the mean from the six simulations as the uncertainty), and the standard deviation was 0.07 kJ mol^{-1}. When using only three λ states for the AWH simulations, the result was 8.59 ± 0.04 kJ mol^{-1}.

The study, from which the starting coordinates and parameters were obtained,[27,55] reported an average hydration free energy from 100 runs of 8.933 ± 0.010 kJ mol^{-1} using BAR and the standard error based on the reported standard deviation from bootstrapping (0.099 kJ mol^{-1}). With MBAR, the results were 8.929 ± 0.009 kJ mol^{-1} with a standard deviation of 0.094 kJ mol^{-1}. We think this difference is due to the change in simulation parameters required when using a more recent GROMACS version (see Sec. II B 1). For comparison, the experimental hydration free energy of methane is 8.08 kJ mol^{-1} at 298.15 K.[60]

B. Ethanol hydration free energy

In Fig. 3 can be seen the average hydration free energies from the different methods. The mean values and the uncertainties are based on five sets of simulations, except when running 1 AWH

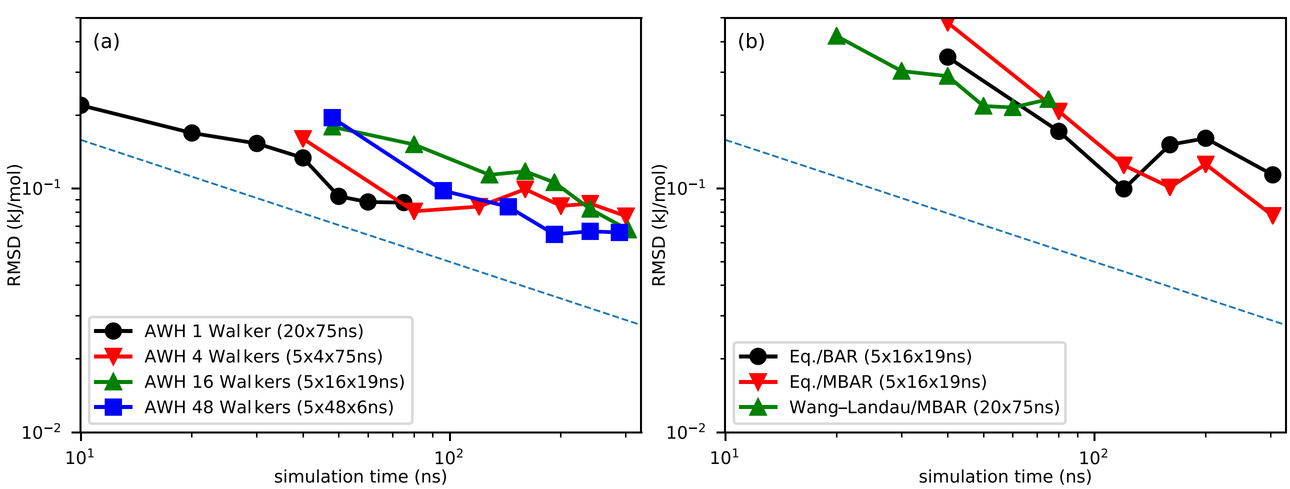

FIG. 4. The root mean square deviation (RMSD) of ethanol hydration free energies to results combined from all data. Provided that the combined results are correct, it is expected that the RMSDs get lower over time. The dashed line shows the theoretical $1/\sqrt{t}$ error evolution. (a) The evolution of the AWH RMSDs over time. (b) The RMSDs of the equilibrium simulations as well as the expanded ensemble Wang–Landau simulations.

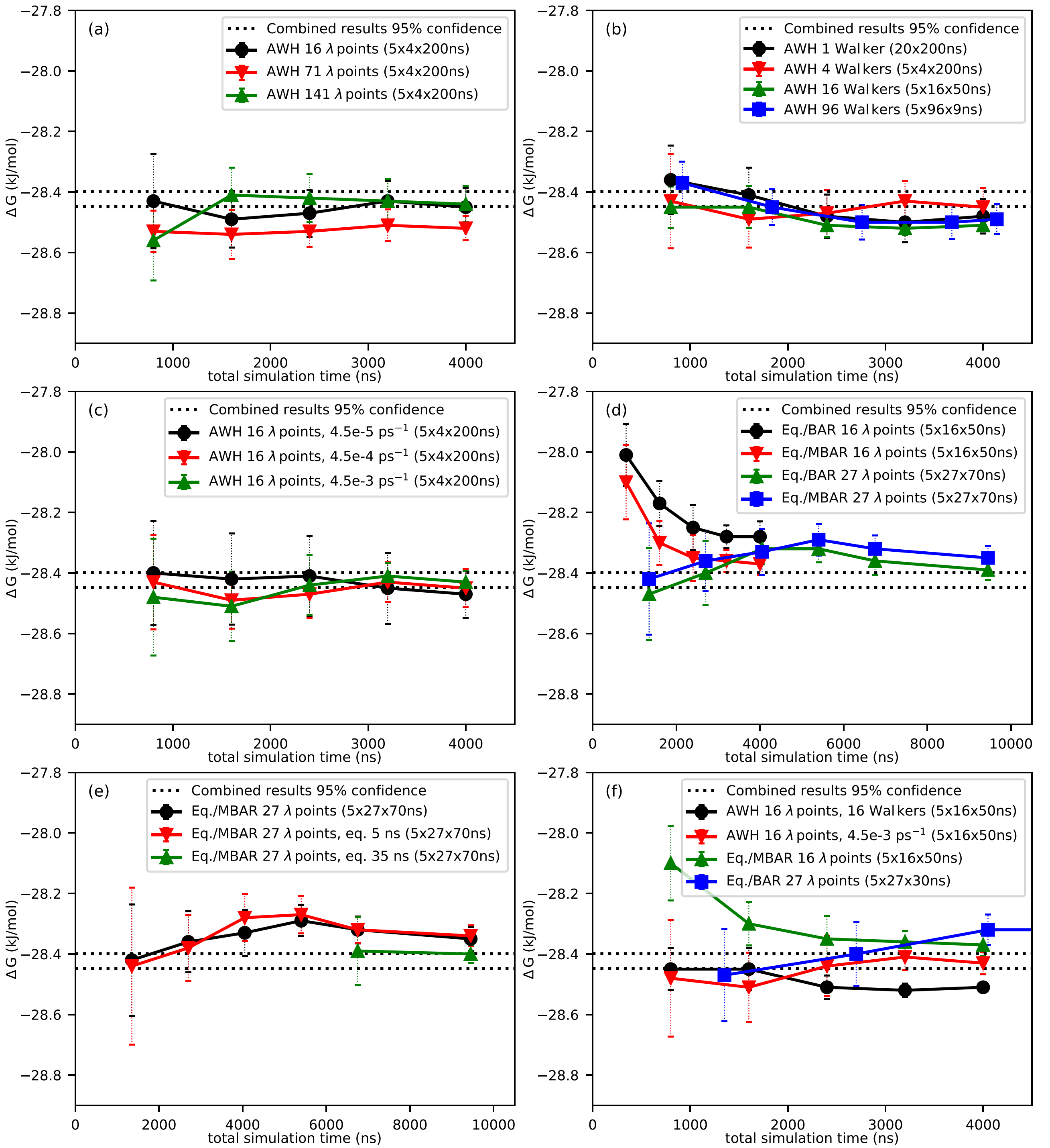

FIG. 5. Convergence of testosterone average hydration free energies using AWH and equilibrium simulations (BAR and MBAR). The plotted time is the total time of all simulations used for calculating the average free energy. (a) AWH simulations with varying number of λ points (four communicating walkers, diffusion constant of 4.5×10^{-4} ps^{-1}). (b) Number of communicating walkers in AWH simulations (16 λ points, diffusion constant of 4.5×10^{-4} ps^{-1}). (c) Varying the AWH diffusion constant input parameter (four communicating walkers). (d) Up to 9450 ns from equilibrium simulations using 16 or 27 λ points analyzed using BAR or MBAR. (e) Comparison of equilibration times for equilibrium simulations with 27 λ points using MBAR, with up to 9450 ns total simulation time. (f) The two datasets with lowest SEM (at 4000 ns) from AWH as well as the one with lowest SEM (at 4000 ns) from equilibrium simulations with 16 and 27 λ points, i.e., MBAR and BAR, respectively. The uncertainties in the plots are 1 SEM.

walker, in which case 20 simulations were used. The 95% confidence interval of the combined time weighted results from the full extent of all simulations was −18.64 to −18.70 kJ mol^{-1} and is shown in Fig. 3. The combined results were the reference used for calculating the RMSD in Fig. 4.

There was no significant difference between the AWH runs after the full simulation time (1500 ns). However, the average hydration free energy from the AWH simulations with one walker, at −18.67 ± 0.02 kJ mol^{-1}, was different compared to the equilibrium simulations, most significantly to MBAR without equilibration, at −18.74 ± 0.02 kJ mol^{-1}, even if the difference was small. This can be observed in Fig. 3(c). The largest difference, disregarding SEM, was between AWH with 16 walkers at −18.63 ± 0.03 kJ mol^{-1} and BAR with 500 ps discarded as equilibration, with a result of −18.78 ± 0.04 kJ mol^{-1}. This difference was also significant. In order to verify that the results were converged, the simulation with 1 AWH walker and the equilibrium simulations were extended to 4000 ns total simulation time (200 ns per AWH simulation and 50 ns per λ state, respectively). After that simulation time, the AWH result was −18.66 ± 0.01 kJ mol^{-1}, and the equilibrium simulations gave −18.78 ± 0.02 kJ mol^{-1} from MBAR analyses with no difference when discarding the first 500 ps. These data are not presented in Fig. 3.

The 20 expanded ensemble[20,52] Wang–Landau simulations,[23] 75 ns each, resulted in a calculated average hydration free energy of −18.62 ± 0.05 kJ mol^{-1}. This was not significantly different compared to the AWH results but to the same equilibrium simulation output as mentioned above.

For comparison, the experimental hydration free energy of ethanol at 298 K is ≈−20.9 kJ mol^{-1}[61,62] and previous MD simulations using the CHARMM[63] forcefield in standard TIP3P[46] water have given −18.87 ± 0.24 kJ mol^{-1}.[47] Those simulations were run with shorter cutoffs than in here and than what is recommended for the CHARMM forcefield in general. The effect of the shortened cutoff, on the Gibbs free energy of solvation of ten compounds, was shown to be ΔΔG − 0.18 ± 0.11 kJ mol^{-1},[47] yielding a corrected result of ∼−18.69 ± 0.26 kJ mol^{-1}, well in agreement with the combined results from the simulations herein.

C. Testosterone hydration free energy

In Fig. 5 can be seen the average hydration free energies from the different methods and settings. The mean values and the uncertainties are based on five sets of simulations, except when running 1 AWH walker, in which case 20 simulations were used. The 95% confidence interval of the combined time weighted results from the full extent of all simulations, also including the output from 35 ns equilibration time discarded of equilibrium simulations with 27 λ points, was −28.40 to −28.45 kJ mol^{-1}, is shown as the dotted lines in Fig. 5 and are used for calculating the RMSD in Fig. 6. To our knowledge, there is no published experimental hydration free energy of testosterone available.

To evaluate the convergence of AWH, and its sensitivity to input parameters, a number of different setups were used, testing the number of communicating walkers, the number of λ points, and the input estimated diffusion constant along the reaction coordinate. The equilibrium simulations were also run using two different λ point distributions. As can be seen in Figs. 5(a)–5(c), after 4000 ns the output from AWH was in the range of (−28.42 ± 0.04) to (−28.52 ± 0.04) kJ mol^{-1}. The only significant differences, in the AWH results, were between 16 λ states, 16 walkers, 4.5×10^{-4} ps^{-1} (hydration free energy −28.51 ± 0.01 kJ mol^{-1}) and 16 λ states, 4 walkers, 4.5×10^{-5} ps^{-1} (hydration free energy −28.42 ± 0.04 kJ mol^{-1}). At 4000 ns, the differences were not significant between the equilibration simulations analyzed using BAR and MBAR. There were significant differences between some results from AWH and from equilibrium simulations at 4000 ns. Extending the equilibrium simulations to 9450 ns and increasing the equilibration times, i.e., the time discarded from analysis, made the results more similar, in general [see Fig. 5(e)]. When discarding 35 ns (out of 70 ns, giving a

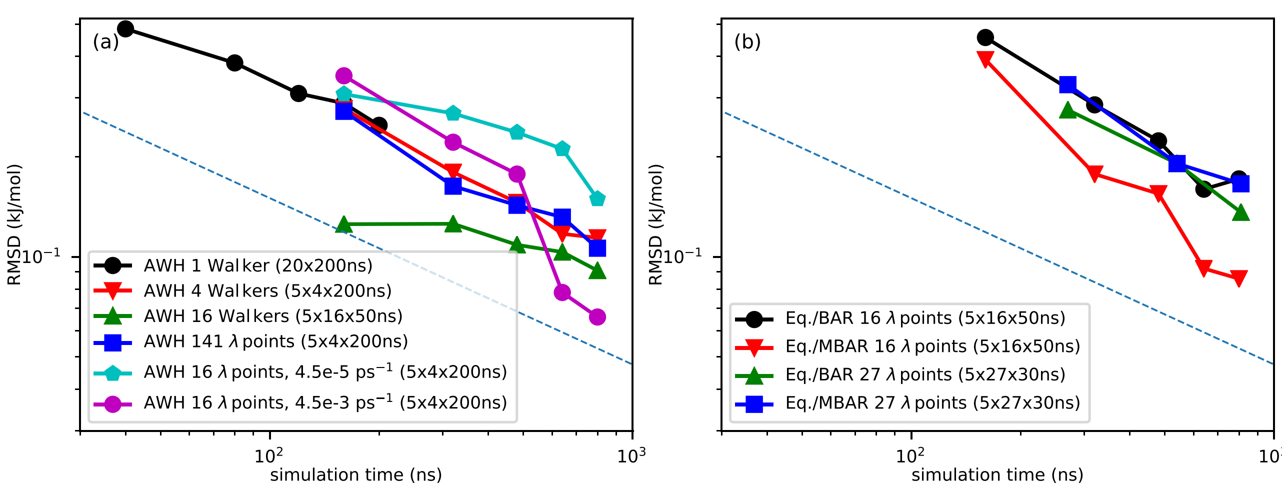

FIG. 6. The root mean square deviation (RMSD) of the testosterone results to the results combined from all data. Provided that the combined results are correct, it is expected that the RMSDs get lower over time. The dashed line shows the theoretical $1/\sqrt{t}$ error evolution. (a) The evolution of the representative AWH results over time. (b) The equilibrium simulations, limited to 4000 ns of the total simulation time.

total simulation time of 9450 ns from five repetitions) per λ window, from the equilibrium simulations, the BAR analysis gave a hydration free energy of -28.46 ± 0.06 kJ mol^{-1} and the MBAR analysis gave -28.40 ± 0.03 kJ mol^{-1}. Comparing these results to those from AWH (from a total simulation time of 4000 ns), the MBAR results were significantly different only compared to the most extreme AWH results, i.e., from 4 walkers with 16 λ states, at -28.51 ± 0.01 kJ mol^{-1} and the BAR results were not different to the AWH output. From Fig. 5(c), it can be deduced that changing the input diffusion coefficient in the range of 4.5×10^{-3} to 4.5×10^{-5} ps^{-1} did not change the results significantly, but the lower the observed standard error of the mean, the higher the input diffusion coefficient (0.04, 0.06, and 0.11 kJ mol^{-1} at 4.5×10^{-3}, 4.5×10^{-4}, and 4.5×10^{-5} ps^{-1}, respectively). However, there were not enough simulations to estimate the uncertainty of the SEM.

Figure 6 shows that approximately one third shorter simulation time was required to reach the same RMSD when using AWH compared to equilibrium simulations. From Fig. 5(e) can be implied that discarding 5 ns in the beginning of the equilibrium simulations might not be enough. At the full total simulation time 9450 ns discarding 35 ns, from each simulation of 70 ns, gave a lower hydration free energy as well as slightly lower SEM than when not discarding any data at all or discarding only 5 ns, despite discarding half of the input data. These differences are not statistically significant but still warrants a closer look at the simulation time lengths, as well as equilibration times, used in many solvation free energy calculation studies. It is possible that much longer simulations, than what are commonly run, might be needed just to reach equilibrium. By comparing the results from the equilibrium simulations with 16 linearly distributed λ points to the optimized distribution of 27 λ states (see the Appendix), in Figs. 5 and 6, it can also be noted that optimizing the distribution of λ points to improve the phase-space overlap between the states does not show any improvement in the results herein. This is surprising as it should improve the efficiency of the sampling.

IV. CONCLUSION AND FUTURE OUTLOOK

AWH is an extended ensemble method to accelerate the calculation of the free energy difference along one or more reaction coordinates.[24,26] In GROMACS, it was originally implemented to use the pull mechanism and thereby steer, e.g., distances or angles between selected groups of atoms.[26]

Herein, we have presented the application of using AWH for alchemical free energy calculations. In all our tests, AWH quickly converges to a stable estimate of the free energy difference. The results are mostly very similar to using BAR or MBAR to analyze equilibrium simulations from separate discrete λ points, as is commonly done, but in some cases, subtle differences are observed. The major one was in the case of ethanol, where the difference between AWH and equilibrium simulations was small (~0.1 kJ mol^{-1}) but statistically significant. A possible reason for the discrepancy is that the equilibrium simulations have difficulties in sampling all the intermediate states adequately. As an additional reference, the same system was simulated using a different expanded ensemble method,[20,52] with an initial Wang–Landau[23] update scheme to determine f_λ, and analyzed using MBAR. The results obtained from this was in agreement with the AWH results but again deviated from the equilibrium simulation results. We thus attribute this discrepancy to insufficient sampling of the equilibrium simulations.

The AWH simulations are not very sensitive to the input parameters and are easy to set up. With the commonly used equilibrium methods, it is important to carefully select the number of λ points and their distribution to achieve efficient sampling.[35–37] With AWH, one can choose a sufficient number of λ values without affecting the efficiency significantly. The flexibility to choose the number of walkers means that it is easy to scale to the available resources and more efficiently parallelize the calculations. As a comparison, when running equilibrium free energy simulations, this may to some extent be restricted by the number of λ states as it may be difficult to efficiently run, e.g., 11 λ states in parallel on 16 nodes of a supercomputer. A disadvantage of AWH, compared to equilibrium methods, is that it is not easy to estimate the uncertainty from one simulation with one or multiple communicating walkers. It is recommended to run multiple simulations, i.e., with independent walkers or sets containing multiple walkers not communicating between the sets, and retrieve the standard deviations from their results, possibly using bootstrapping or similar alternatives. Even for equilibrium simulations, this is a more reliable way to estimate the standard deviations, especially if there might be states that are insufficiently sampled.

Equilibrium simulations analyzed using BAR and MBAR are efficient for calculating the solvation free energy of noncomplex molecules.[27] As long as the sampling of the intermediate states is relatively efficient, it should not be expected that AWH will perform much better. Both approaches use reweighting techniques to optimize the use of the sampled data. On the other hand, if the sampling is challenging, for example, in cases where there are states with long correlation times, extended ensemble methods, such as AWH, can be expected to perform significantly better.[64,65] Compared to the equilibrium simulations, the AWH results fluctuate less over time. The reported standard errors from AWH simulations are, in general, lower than from equilibrium simulations of the same total length. It would be conservative to state that the convergence obtained using AWH is at least as fast as when using conventional methods. In the testosterone case, a reduced simulation time of approximately one third was required to reach the same RMSD when using AWH. There is room for further improvement by optimizing the target distribution (see Sec. I A 2). What is important to highlight is the wide applicability of the method. The flexibility of the AWH algorithm also means that it is easy to run multidimensional reaction coordinates, which is enabled in the current GROMACS implementation. This makes it possible to combine alchemical free energy calculations with, e.g., a pull reaction coordinate. This can be used for more advanced binding free energy calculations or membrane permeability simulations, where the possibility to decouple interactions can make it more easy to escape local free energy minima and regions of very slow diffusion. One example of potential use is improving the transfer free energy calculations between a solvent and the gel-like intercellular lipid barrier in the human stratum corneum and to also significantly advance the permeability coefficient calculations through the same system.[66,67]

TABLE II. Examples of λ point distribution used in the simulations. 0.00 means that the molecule is fully decoupled (does not interact with its surroundings), and 1.00 means that the interactions are turned on. The van der Waals interactions are listed in the vdW columns and the Coulomb interactions in the C columns.

16 λ		21 λ		Optimized 27 λ	
vdW	C	vdW	C	vdW	C
0.00	0.00	0.00	0.00	0.00	0.00
0.10	0.00	0.10	0.00	0.12	0.00
0.20	0.00	0.20	0.00	0.16	0.00
0.30	0.00	0.30	0.00	0.20	0.00
0.40	0.00	0.40	0.00	0.22	0.00
0.50	0.00	0.50	0.00	0.24	0.00
0.60	0.00	0.60	0.00	0.26	0.00
0.70	0.00	0.70	0.00	0.28	0.00
0.80	0.00	0.80	0.00	0.30	0.00
0.90	0.00	0.90	0.00	0.33	0.00
1.00	0.00	1.00	0.00	0.36	0.00
1.00	0.20	1.00	0.10	0.40	0.00
1.00	0.40	1.00	0.20	0.45	0.00
1.00	0.60	1.00	0.30	0.50	0.00
1.00	0.80	1.00	0.40	0.56	0.00
1.00	1.00	1.00	0.50	0.63	0.00
		1.00	0.60	0.71	0.00
		1.00	0.70	0.80	0.00
		1.00	0.80	0.90	0.00
		1.00	0.90	1.00	0.00
		1.00	1.00	1.00	0.17
				1.00	0.34
				1.00	0.48
				1.00	0.62
				1.00	0.75
				1.00	0.88
				1.00	1.00

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APPENDIX: λ POINT DISTRIBUTION

Table II shows the examples of λ point distribution used in the simulations.

DATA AVAILABILITY

The data that support the findings of this study are openly available at https://doi.org/10.5281/zenodo.4592004.

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